

EXERCISE 13

Detection of real-time threats, analyse recorded traffic files and identify anomalies

Aim: To learn how to use Snort for real-time threat detection, traffic analysis, and anomaly identification using both live and recorded network data.

The screenshot shows the Snort exercise interface. At the top, there's a header with the Snort logo and the text "Learn how to use Snort to detect real-time threats, analyse recorded traffic files and identify anomalies." Below this, there's a progress bar indicating "Room completed (100%)". A list of 11 tasks is displayed, each with a green checkmark icon and a dropdown arrow. The tasks are:

- Task 1: Introduction
- Task 2: Interactive Material and VM
- Task 3: Introduction to IDS/IPS
- Task 4: First Interaction with Snort
- Task 5: Operation Mode 1: Sniffer Mode
- Task 6: Operation Mode 2: Pocket Logger Mode
- Task 7: Operation Mode 3: IDS/IPS
- Task 8: Operation Mode 4: PCAP Investigation
- Task 9: Snort Rule Structure
- Task 10: Snort2 Operation Logic: Points to Remember
- Task 11: Conclusion

The screenshot shows the answer section for the first four tasks. Each task has a text input field, a "Correct Answer" button, and a "Hint" button.

- Task 1: "Navigate to the Task-Exercises folder and run the command `./easy.sh` and write the output." The answer is "Too Easy!".
- Task 2: "Run the Snort instance and check the build number." The answer is "149".
- Task 3: "Test the current instance with `/etc/snort/snort.conf` file and check how many rules are loaded with the current build." The answer is "4151".
- Task 4: "Test the current instance with `/etc/snort/snortv2.conf` file and check how many rules are loaded with the current build." The answer is "1".

Task 5: "What is the number of the detected HTTP GET methods?" The answer is "2".

Which IDS or IPS type can help you stop the threats on a local machine?

HIPS

✓ Correct Answer

Which IDS or IPS type can help you detect threats on a local network?

NIDS

✓ Correct Answer

Which IDS or IPS type can help you detect the threats on a local machine?

HIDS

✓ Correct Answer

Which IDS or IPS type can help you stop the threats on a local network?

NIPS

✓ Correct Answer

Which described solution works by detecting anomalies in the network?

NBA

✓ Correct Answer

According to the official description of the snort, what kind of NIPS is it?

full-blown

✓ Correct Answer

NBA training period is also known as ...

baselining

✓ Correct Answer

Investigate the traffic with the default configuration file **with ASCII mode**.

```
sudo snort -dev -K ASCII -l .
```

Execute the traffic generator script and choose "TASK-6 Exercise". Wait until the traffic ends, then stop the Snort instance. Now analyse the output summary and answer the question.

```
sudo ./traffic-generator.sh
```

Now, you should have the logs in the current directory. Navigate to folder "145.254.180.233". What is the source port used to connect port 53?

3003

✓ Correct Answer

🔗 Hint

Use `snort.log.1640048004`

Read the `snort.log` file with Snort; what is the IP ID of the 10th packet?

```
snort -r snort.log.1640048004 -n 10
```

49313

✓ Correct Answer

🔗 Hint

Read the "`snort.log.1640048004`" file with Snort; what is the referer of the 4th packet?

<http://www.ethereal.com/development.html>

✓ Correct Answer

🔗 Hint

Read the "`snort.log.1640048004`" file with Snort; what is the Ack number of the 8th packet?

0x3BAFFFF3

✓ Correct Answer

Read the "`snort.log.1640048004`" file with Snort; what is the number of the "TCP port 80" packets?

41

✓ Correct Answer

🔗 Hint

Investigate the `mx-1.pcap` file with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l - -r mx-1.pcap
```

What is the number of the generated alerts?

170 ✓ Correct Answer

Keep reading the output. How many TCP Segments are Queued?

18 ✓ Correct Answer

Keep reading the output. How many "HTTP response headers" were extracted?

3 ✓ Correct Answer

Investigate the `mx-1.pcap` file with the second configuration file.

```
sudo snort -c /etc/snort/snortv2.conf -A full -l - -r mx-1.pcap
```

What is the number of the generated alerts?

68 ✓ Correct Answer

Investigate the `mx-2.pcap` file with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l - -r mx-2.pcap
```

What is the number of the generated alerts?

340 ✓ Correct Answer 🔍 Hint

Keep reading the output. What is the number of the detected TCP packets?

82 ✓ Correct Answer

Investigate the `mx-2.pcap` and `mx-3.pcap` files with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l - --pcap-list="mx-2.pcap mx-3.pcap"
```

What is the number of the generated alerts?

1920 ✓ Correct Answer

Use "task3.pcap". Write a rule to filter IP ID "35368" and run it against the given pcap file. What is the request name of the detected packet? You may use this command: "snort -c local.rules -A full -l - -r task3.pcap"

TIME STAMP REQUEST ✓ Correct Answer 🔍 Hint

Clear the previous alert file and comment out the old rules. Create a rule to filter packets with `Syn` flag and run it against the given pcap file. What is the number of detected packets?

1 ✓ Correct Answer

Clear the previous alert file and comment out the old rules. Write a rule to filter packets with `Push-Ack` flags and run it against the given pcap file. What is the number of detected packets?

216 ✓ Correct Answer

Clear the previous alert file and comment out the old rules. Create a rule to filter `UDP` packets with the same source and destination IP and run it against the given pcap file. What is the number of packets that show the same source and destination address?

7 ✓ Correct Answer

Case Example - An analyst modified an existing rule successfully. Which rule option must the analyst change after the implementation?

rev ✓ Correct Answer

Result: Successfully configured Snort to detect live threats, analyzed pcap traffic files, and identified suspicious patterns and anomalies in network behavior.